

```
import pandas as pd
import numpy as np
import re
import nltk

from sklearn.model_selection import train_test_split
from sklearn.feature_extraction.text import TfidfVectorizer
from sklearn.linear_model import LogisticRegression
from sklearn.metrics import accuracy_score, confusion_matrix, classification_report
```

Load Datasets

```
true_df = pd.read_csv('/content/True (1).csv')
fake_df = pd.read_csv('/content/Fake (1).csv')
manual_df = pd.read_csv('/content/manual_testing.csv')
```

```
true_df.head()
```

	title	text	subject	date
0	As U.S. budget fight looms, Republicans flip t...	WASHINGTON (Reuters) - The head of a conservat...	politicsNews	December 31, 2017
1	U.S. military to accept transgender recruits o...	WASHINGTON (Reuters) - Transgender people will...	politicsNews	December 29, 2017
2	Senior U.S. Republican senator: 'Let Mr. Muell...	WASHINGTON (Reuters) - The special counsel inv...	politicsNews	December 31, 2017
3	FBI Russia probe helped by Australian diplomat...	WASHINGTON (Reuters) - Trump campaign adviser ...	politicsNews	December 30, 2017
4	Trump wants Postal Service to charge 'much mor...	SEATTLE/WASHINGTON (Reuters) - President Donal...	politicsNews	December 29, 2017

```
fake_df.head()
```

	title	text	subject	date
0	Donald Trump Sends Out Embarrassing New Year'...	Donald Trump just couldn t wish all Americans ...	News	December 31, 2017
1	Drunk Bragging Trump Staffer Started Russian ...	House Intelligence Committee Chairman Devin Nu...	News	December 31, 2017
2	Sheriff David Clarke Becomes An Internet Joke...	On Friday, it was revealed that former Milwauk...	News	December 30, 2017
3	Trump Is So Obsessed He Even Has Obama's Name...	On Christmas day, Donald Trump announced that ...	News	December 29, 2017
4	Pope Francis Just Called Out Donald Trump Dur...	Pope Francis used his annual Christmas Day mes...	News	December 25, 2017

```
manual_df.head()
```

	Unnamed: 0	title	text	subject	date	class
0	23471	Seven Iranians freed in the prisoner swap have...	21st Century Wire says This week, the historic...	Middle-east	January 20, 2016	0
1	23472	#Hashtag Hell & The Fake Left	By Dady Chery and Gilbert MercierAll writers ...	Middle-east	January 19, 2016	0
2	23473	Astroturfing: Journalist Reveals Brainwashing ...	Vic Bishop Waking TimesOur reality is carefull...	Middle-east	January 19, 2016	0

```
fake_df.shape, true_df.shape,manual_df.shape
```

```
((23481, 4), (21417, 4), (20, 6))
```

```
fake_df.columns,true_df.columns,manual_df.columns
```

```
(Index(['title', 'text', 'subject', 'date'], dtype='object'),
Index(['title', 'text', 'subject', 'date'], dtype='object'),
Index(['Unnamed: 0', 'title', 'text', 'subject', 'date', 'class'], dtype='object'))
```

```
fake_df.isnull().sum()
```

```

      0
title    0
text     0
subject  0
date     0

dtype: int64

```

```
true_df.isnull().sum()
```

```

      0
title    0
text     0
subject  0
date     0

dtype: int64

```

```
manual_df.isnull().sum()
```

```

      0
Unnamed: 0  0
title      0
text       0
subject    0
date       0
class      0

dtype: int64

```

✓ Add Class Label to Fake Dataset

```

fake_df["class"] = 0 # Fake News
true_df["class"] = 1 # True News

```

✓ Combine True & Fake Datasets

```

news_df = pd.concat([true_df, fake_df], axis=0)
news_df.reset_index(drop=True, inplace=True)
print(news_df)

```

```

      title \
0  As U.S. budget fight looms, Republicans flip t...
1  U.S. military to accept transgender recruits o...
2  Senior U.S. Republican senator: 'Let Mr. Muell...
3  FBI Russia probe helped by Australian diplomat...
4  Trump wants Postal Service to charge 'much mor...
...
44893  McPain: John McCain Furious That Iran Treated ...
44894  JUSTICE? Yahoo Settles E-mail Privacy Class-ac...
44895  Sunnistan: US and Allied 'Safe Zone' Plan to T...
44896  How to Blow $700 Million: Al Jazeera America F...
44897  10 U.S. Navy Sailors Held by Iranian Military ...

      text      subject \
0  WASHINGTON (Reuters) - The head of a conservat...  politicsNews
1  WASHINGTON (Reuters) - Transgender people will...  politicsNews
2  WASHINGTON (Reuters) - The special counsel inv...  politicsNews

```

```

3      WASHINGTON (Reuters) - Trump campaign adviser ... politicsNews
4      SEATTLE/WASHINGTON (Reuters) - President Donal... politicsNews
...
44893  21st Century Wire says As 21WIRE reported earl... Middle-east
44894  21st Century Wire says It s a familiar theme. ... Middle-east
44895  Patrick Henningsen 21st Century WireRemember ... Middle-east
44896  21st Century Wire says Al Jazeera America will... Middle-east
44897  21st Century Wire says As 21WIRE predicted in ... Middle-east

      date  class
0      December 31, 2017      1
1      December 29, 2017      1
2      December 31, 2017      1
3      December 30, 2017      1
4      December 29, 2017      1
...
44893  January 16, 2016      0
44894  January 16, 2016      0
44895  January 15, 2016      0
44896  January 14, 2016      0
44897  January 12, 2016      0

```

[44898 rows x 5 columns]

Data Cleaning (Text Preprocessing)

Define Cleaning Function

```

def clean_text(text):
    text = text.lower()
    text = re.sub(r'\W', ' ', text)
    text = re.sub(r'\s+', ' ', text)
    return text

```

Apply Cleaning

```

news_df["text"] = news_df["text"].apply(clean_text)
manual_df["text"] = manual_df["text"].apply(clean_text)
print(news_df)
print(manual_df)

```

```

      title \
0      As U.S. budget fight looms, Republicans flip t...
1      U.S. military to accept transgender recruits o...
2      Senior U.S. Republican senator: 'Let Mr. Muell...
3      FBI Russia probe helped by Australian diplomat...
4      Trump wants Postal Service to charge 'much mor...
...
44893  McPain: John McCain Furious That Iran Treated ...
44894  JUSTICE? Yahoo Settles E-mail Privacy Class-ac...
44895  Sunnistan: US and Allied 'Safe Zone' Plan to T...
44896  How to Blow $700 Million: Al Jazeera America F...
44897  10 U.S. Navy Sailors Held by Iranian Military ...

      text      subject \
0      washington reuters the head of a conservative ... politicsNews
1      washington reuters transgender people will be ... politicsNews
2      washington reuters the special counsel investi... politicsNews
3      washington reuters trump campaign adviser geor... politicsNews
4      seattle washington reuters president donald tr... politicsNews
...
44893  21st century wire says as 21wire reported earl... Middle-east
44894  21st century wire says it s a familiar theme w... Middle-east
44895  patrick henningsen 21st century wireremember w... Middle-east
44896  21st century wire says al jazeera america will... Middle-east
44897  21st century wire says as 21wire predicted in ... Middle-east

      date  class
0      December 31, 2017      1
1      December 29, 2017      1
2      December 31, 2017      1
3      December 30, 2017      1
4      December 29, 2017      1
...
44893  January 16, 2016      0

```

```

44894 January 16, 2016 0
44895 January 15, 2016 0
44896 January 14, 2016 0
44897 January 12, 2016 0

[44898 rows x 5 columns]
   Unnamed: 0  title \
0      23471  Seven Iranians freed in the prisoner swap have...
1      23472  #Hashtag Hell & The Fake Left
2      23473  Astroturfing: Journalist Reveals Brainwashing ...
3      23474  The New American Century: An Era of Fraud
4      23475  Hillary Clinton: 'Israel First' (and no peace ...
5      23476  McPain: John McCain Furious That Iran Treated ...
6      23477  JUSTICE? Yahoo Settles E-mail Privacy Class-ac...
7      23478  Sunnistan: US and Allied 'Safe Zone' Plan to T...
8      23479  How to Blow $700 Million: Al Jazeera America F...
9      23480  10 U.S. Navy Sailors Held by Iranian Military ...
10     21407  Mata Pires, owner of embattled Brazil builder ...
11     21408  U.S., North Korea clash at U.N. forum over nuc...
12     21409  U.S., North Korea clash at U.N. arms forum on ...
13     21410  Headless torso could belong to submarine journ...
14     21411  North Korea shipments to Syria chemical arms a...
15     21412  'Fully committed' NATO backs new U.S. approach...

```

Feature Selection

- ✓ We will use only text (best for fake news detection).

```

X = news_df["text"]
y = news_df["class"]

```

- ✓ Train-Test Split

```

X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.25, random_state=42
)

```

- ✓ Convert Text to Numbers (TF-IDF)

```

vectorizer = TfidfVectorizer(stop_words='english', max_df=0.7)
X_train_tfidf = vectorizer.fit_transform(X_train)
X_test_tfidf = vectorizer.transform(X_test)

```

- ✓ Train AI Model (Logistic Regression)

```

model = LogisticRegression()
model.fit(X_train_tfidf, y_train)

```

▼ LogisticRegression ⓘ ?

```

LogisticRegression()

```

```
model.coef_
```

```

array([[ -0.61093631,  0.37510677,  0.07845906, ..., -0.0053647 ,
        -0.00121986,  0.00726873]])

```

- ✓ Model Evaluation

```

y_pred = model.predict(X_test_tfidf)

log_reg_acc=print("Accuracy:", accuracy_score(y_test, y_pred))
print("\nConfusion Matrix:\n", confusion_matrix(y_test, y_pred))
print("\nClassification Report:\n", classification_report(y_test, y_pred))

```

Accuracy: 0.9850334075723831

Confusion Matrix:

```

[[5725  104]
 [   64 5332]]

```

Classification Report:

	precision	recall	f1-score	support
0	0.99	0.98	0.99	5829
1	0.98	0.99	0.98	5396
accuracy			0.99	11225
macro avg	0.98	0.99	0.99	11225
weighted avg	0.99	0.99	0.99	11225

Manual Testing Dataset (Your 3rd Dataset)

▼ Prepare Manual Test Data

```

manual_text = vectorizer.transform(manual_df["text"])
manual_pred = model.predict(manual_text)
print(manual_pred)

```

```
[0 0 0 0 0 0 0 0 0 0 1 1 1 1 1 1 1 1 1]
```

▼ Compare Predictions

```

manual_df["Predicted_Class"] = manual_pred
print(manual_df[["text", "class", "Predicted_Class"]])

```

	text	class	Predicted_Class
0	21st century wire says this week the historic ...	0	0
1	by dady chery and gilbert mercierall writers ...	0	0
2	vic bishop waking timesour reality is carefull...	0	0
3	paul craig robertsin the last years of the 20t...	0	0
4	robert fantina counterpunchalthough the united...	0	0
5	21st century wire says as 21wire reported earl...	0	0
6	21st century wire says it s a familiar theme w...	0	0
7	patrick henningsen 21st century wireremember w...	0	0
8	21st century wire says al jazeera america will...	0	0
9	21st century wire says as 21wire predicted in ...	0	0
10	sao paulo reuters cesar mata pires the owner a...	1	1
11	geneva reuters north korea and the united stat...	1	1
12	geneva reuters north korea and the united stat...	1	1
13	copenhagen reuters danish police said on tuesd...	1	1
14	united nations reuters two north korean shipme...	1	1
15	brussels reuters nato allies on tuesday welcom...	1	1
16	london reuters lexisnexis a provider of legal ...	1	1
17	minsk reuters in the shadow of disused soviet ...	1	1
18	moscow reuters vatican secretary of state card...	1	1
19	jakarta reuters indonesia will buy 11 sukhoi f...	1	1

▼ Accuracy on Manual Testing

```

print("Manual Test Accuracy:",
      accuracy_score(manual_df["class"], manual_pred))

```

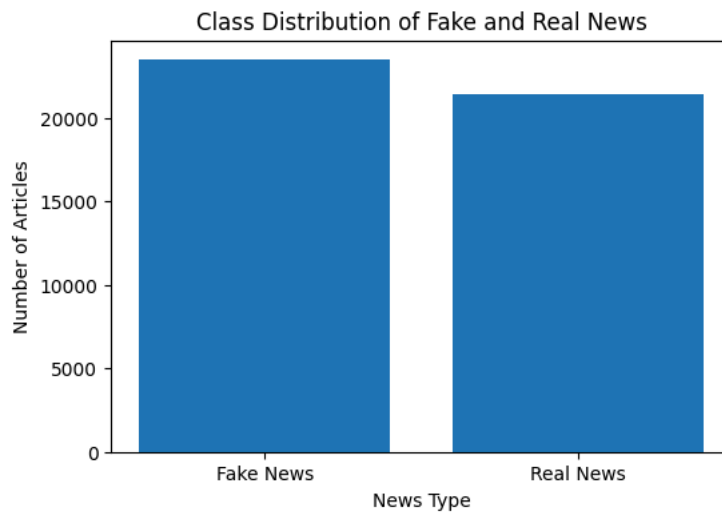
Manual Test Accuracy: 1.0

GRAPH 1: Class Distribution (Fake vs Real News)

```
import matplotlib.pyplot as plt

# Count class values
class_counts = news_df["class"].value_counts()

plt.figure(figsize=(6,4))
plt.bar(["Fake News", "Real News"], class_counts)
plt.xlabel("News Type")
plt.ylabel("Number of Articles")
plt.title("Class Distribution of Fake and Real News")
plt.show()
```

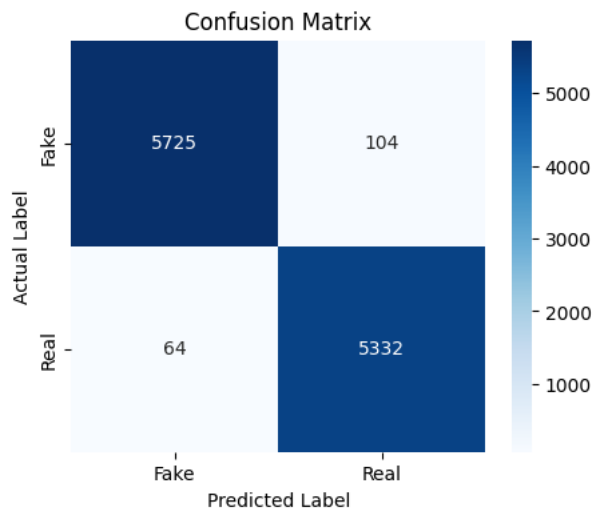


GRAPH 2: Confusion Matrix

```
from sklearn.metrics import confusion_matrix
import seaborn as sns

cm = confusion_matrix(y_test, y_pred)

plt.figure(figsize=(5,4))
sns.heatmap(cm, annot=True, fmt="d", cmap="Blues",
            xticklabels=["Fake", "Real"],
            yticklabels=["Fake", "Real"])
plt.xlabel("Predicted Label")
plt.ylabel("Actual Label")
plt.title("Confusion Matrix")
plt.show()
```

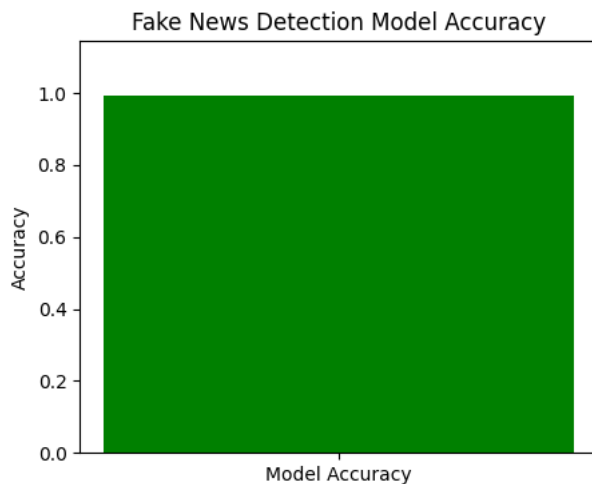


GRAPH 3: Model Accuracy

```
from sklearn.metrics import accuracy_score

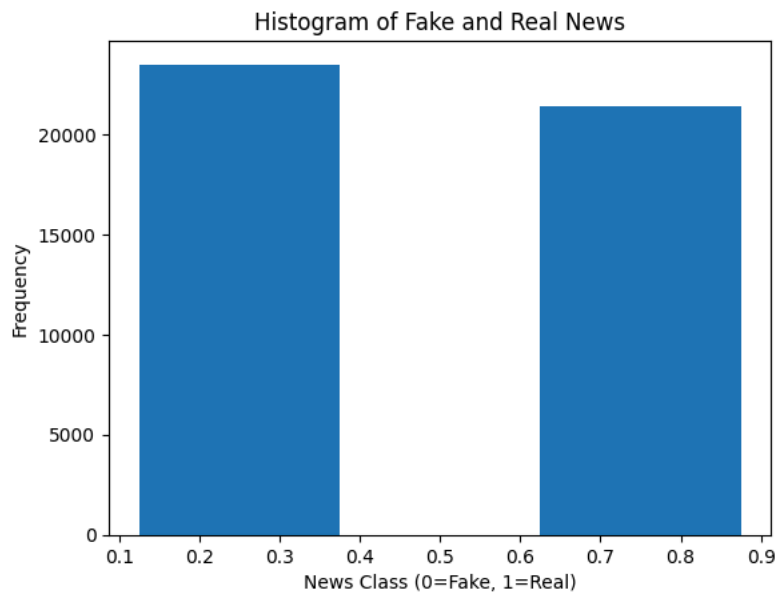
accuracy = accuracy_score(y_test, y_pred)

plt.figure(figsize=(5,4))
plt.bar(["Model Accuracy"], [accuracy], color="green")
plt.ylim(0, max(accuracies) + 0.15)
plt.ylabel("Accuracy")
plt.title("Fake News Detection Model Accuracy")
plt.show()
```



GRAPH 4: Histogram of Fake and Real News

```
plt.hist(news_df['class'], bins=2, rwidth=0.5)
plt.xlabel("News Class (0=Fake, 1=Real)")
plt.ylabel("Frequency")
plt.title("Histogram of Fake and Real News")
plt.show()
```



GRAPH 5:Manual Testing Prediction Results

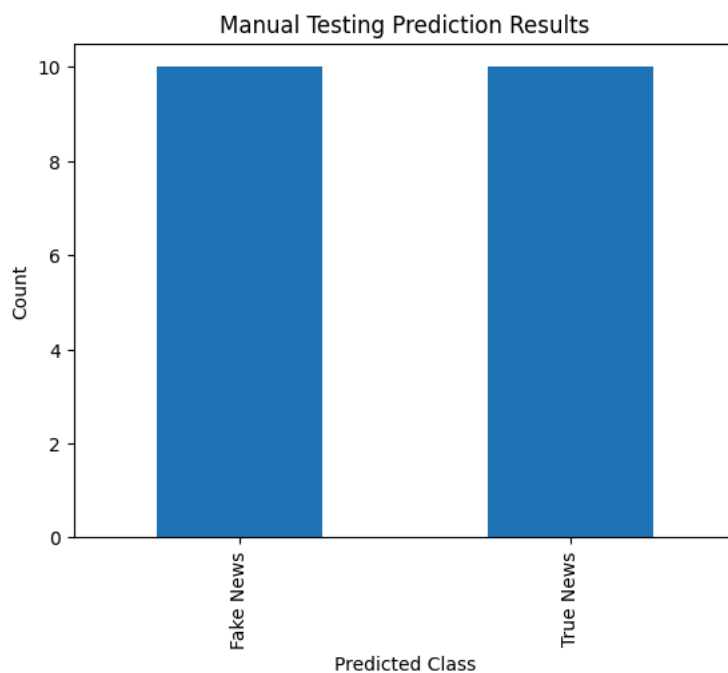
```
import matplotlib.pyplot as plt

# Transform text using trained vectorizer
manul_vec = vectorizer.transform(manul_df['text'])

# Predict using trained model
manul_df['Predicted_Class'] = model.predict(manul_vec)

# Bar graph of predictions
manul_df['Predicted_Class'].value_counts().rename(
    index={0: 'Fake News', 1: 'True News'})
).plot(kind='bar')

plt.xlabel("Predicted Class")
plt.ylabel("Count")
plt.title("Manual Testing Prediction Results")
plt.show()
```

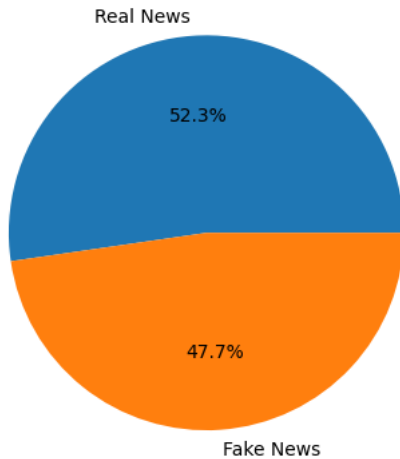


Graph 6: Pie Chart of Fake vs Real News Distribution

```
counts = news_df['class'].value_counts()

# Pie chart
plt.pie(counts, labels=['Real News', 'Fake News'], autopct='%1.1f%%')
plt.title("Pie Chart of Fake vs Real News Distribution")
plt.show()
```

Pie Chart of Fake vs Real News Distribution



METHOD 1: SVM Model Training

```
from sklearn.svm import LinearSVC

svm_model = LinearSVC()
svm_model.fit(X_train_tfidf, y_train)
```

LinearSVC ⓘ ?

LinearSVC()

Final prediction result using SVM METHOD

```
y_pred = svm_model.predict(X_test_tfidf)
```

Model Evaluation using SVM METHOD

```
from sklearn.metrics import accuracy_score, classification_report, confusion_matrix

svm_acc=print("Accuracy:", accuracy_score(y_test, y_pred))
cmsvm=print(confusion_matrix(y_test, y_pred))
print(classification_report(y_test, y_pred))
```

```
Accuracy: 0.9937639198218263
[[5785  44]
 [ 26 5370]]
           precision    recall  f1-score   support

0             1.00        0.99        0.99        5829
1             0.99        1.00        0.99        5396
```

accuracy			0.99	11225
macro avg	0.99	0.99	0.99	11225
weighted avg	0.99	0.99	0.99	11225

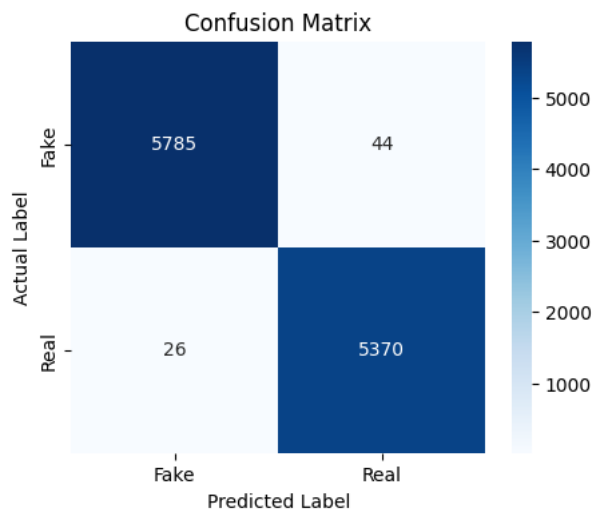
```

from sklearn.metrics import confusion_matrix
import seaborn as sns

cmsvm = confusion_matrix(y_test, y_pred)

plt.figure(figsize=(5,4))
sns.heatmap(cmsvm, annot=True, fmt="d", cmap="Blues",
            xticklabels=["Fake", "Real"],
            yticklabels=["Fake", "Real"])
plt.xlabel("Predicted Label")
plt.ylabel("Actual Label")
plt.title("Confusion Matrix")
plt.show()

```



▼ METHOD 2: Random Forest Classifier Code

```
from sklearn.ensemble import RandomForestClassifier
```

▼ Train Model

```

rf_model = RandomForestClassifier(
    n_estimators=100,
    random_state=42
)
rf_model.fit(X_train_tfidf, y_train)

```

▼ RandomForestClassifier ⓘ ?

RandomForestClassifier(random_state=42)

▼ Prediction

```
rf_pred = rf_model.predict(X_test_tfidf)
```

▼ Evaluation

```
from sklearn.metrics import accuracy_score, classification_report

rf_acc=print("Random Forest Accuracy:", accuracy_score(y_test, rf_pred))
print(classification_report(y_test, rf_pred))
cmrf=print("confusion_matrix:",confusion_matrix(y_test, y_pred))
```

```
Random Forest Accuracy: 0.9908240534521158
      precision    recall  f1-score   support

      0       0.99      0.99      0.99     5829
      1       0.99      0.99      0.99     5396

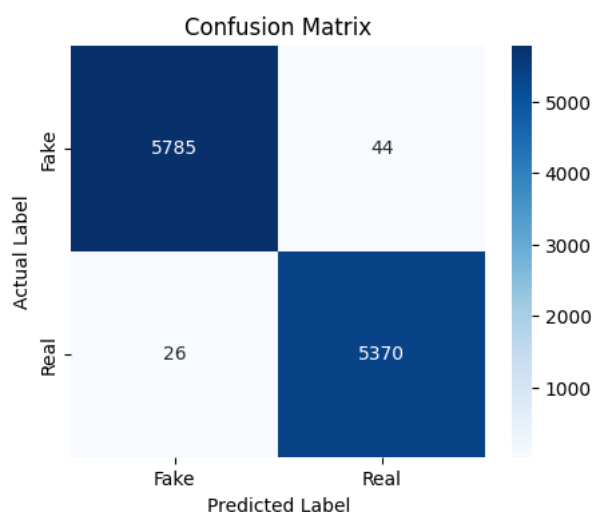
   accuracy          0.99          11225
  macro avg       0.99      0.99      0.99     11225
 weighted avg       0.99      0.99      0.99     11225

confusion_matrix: [[5785   44]
 [  26 5370]]
```

```
from sklearn.metrics import confusion_matrix
import seaborn as sns

cmrf = confusion_matrix(y_test, y_pred)

plt.figure(figsize=(5,4))
sns.heatmap(cmrf, annot=True, fmt="d", cmap="Blues",
            xticklabels=["Fake", "Real"],
            yticklabels=["Fake", "Real"])
plt.xlabel("Predicted Label")
plt.ylabel("Actual Label")
plt.title("Confusion Matrix")
plt.show()
```



Model 3:KNN

```
from sklearn.neighbors import KNeighborsClassifier
from sklearn.metrics import accuracy_score, confusion_matrix, classification_report

#Create KNN model
knn_model = KNeighborsClassifier(
    n_neighbors=5,
    metric='cosine' # works better with TF-IDF
)

#Train (fit) the model
knn_model.fit(X_train_tfidf, y_train)

#Make predictions
knn_pred = knn_model.predict(X_test_tfidf)

#Evaluate the model
```

```

knn_accuracy = accuracy_score(y_test, knn_pred)

print("KNN Accuracy:", knn_accuracy)
cmmknn=print("\nConfusion Matrix:\n", confusion_matrix(y_test, knn_pred))
print("\nClassification Report:\n", classification_report(y_test, knn_pred))

```

KNN Accuracy: 0.8602227171492205

Confusion Matrix:

```

[[4662 1167]
 [ 402 4994]]

```

Classification Report:

	precision	recall	f1-score	support
0	0.92	0.80	0.86	5829
1	0.81	0.93	0.86	5396
accuracy			0.86	11225
macro avg	0.87	0.86	0.86	11225
weighted avg	0.87	0.86	0.86	11225

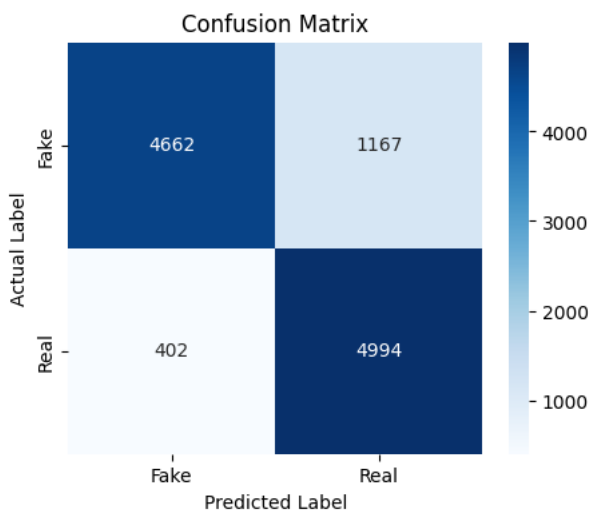
```

from sklearn.metrics import confusion_matrix
import seaborn as sns

cmmknn = confusion_matrix(y_test, knn_pred)

plt.figure(figsize=(5,4))
sns.heatmap(cmmknn, annot=True, fmt="d", cmap="Blues",
            xticklabels=["Fake", "Real"],
            yticklabels=["Fake", "Real"])
plt.xlabel("Predicted Label")
plt.ylabel("Actual Label")
plt.title("Confusion Matrix")
plt.show()

```



✓ #COMPARE 3 MODELS ACCURACY

```

import matplotlib.pyplot as plt
from sklearn.metrics import accuracy_score

# Recalculate accuracies since previous assignments were incorrect (assigned None from print function)

# Logistic Regression Accuracy
y_pred_log_reg = model.predict(X_test_tfidf)
log_reg_acc = accuracy_score(y_test, y_pred_log_reg)

# SVM Accuracy
y_pred_svm = svm_model.predict(X_test_tfidf)

```

```

svm_acc = accuracy_score(y_test, y_pred_svm)

# Random Forest Accuracy (rf_pred is already available from previous execution)
rf_acc = accuracy_score(y_test, rf_pred)

#KNN
knn_pred = knn_model.predict(X_test_tfidf)
knn_accuracy = accuracy_score(y_test, knn_pred)

models = ['Logistic Regression', 'SVM', 'Random Forest', 'KNN']
accuracies = [log_reg_acc, svm_acc, rf_acc, knn_accuracy]

# Plot bar graph
plt.figure(figsize=(9, 5))
bars = plt.bar(models, accuracies, color=['skyblue', 'lightgreen', 'salmon', 'darkblue'])

# Add accuracy values on top of bars
for bar in bars:
    height = bar.get_height()
    plt.text(bar.get_x() + bar.get_width()/2, height,
             f'{height:.4f}', ha='center', va='bottom')

plt.xlabel('Models')
plt.ylabel('Accuracy')
plt.title('Accuracy Comparison of ML Models')
plt.ylim(0, max(accuracies) + 0.15)
plt.grid(axis='y', linestyle='--', alpha=0.7)

```

